



CS6502 Applied Big Data and Visualisation

Group 4:

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I. Abstract

Social media usage is highly prevalent in the modern world and has become an integral part of daily life. As its rapid growth continues, concerns about its impact on mental health, productivity, and overall well-being have also increased, with excessive use often leading to social media addiction. In light of that, this project takes a regression-based approach to predict how active or engaged a user is on the social media platform Instagram based on demographic information, lifestyle behaviours, and platform-based interactions. The objective is to see how accurately the *user_engagement_score* can be predicted and to determine which of the 58 features have the strongest influence on it. This study utilises a large synthetic dataset from Kaggle with around 1.5 million rows and 58 features to train and evaluate multiple models, including Multiple Linear Regression (MLR), Decision Tree (DT), Random Forest (RF), and Gradient Boosting Trees (GBT). All models achieved strong predictive performance, with the best performing model (GBT) achieving the lowest RMSE and highest R^2 score among all models. In addition, this project also includes insightful visualisations that highlight interesting relationships and patterns, providing a clearer understanding of the data. Finally, based on the RF feature importance results, the most influential factor is *sessions_per_day*, while the least important is *perceived_stress_score*.

II. Introduction

As previously stated, we live in a digital age where social media has fundamentally changed how we communicate and manage relationships. Usage has become so normalised that many people do not realise they have fallen into behavioural addiction. And

even though social media helps us connect with our friends or raise awareness on important issues, overusing it for sure weighs down our mental well-being and happiness. For example, according to a recent study by [1], heavy use of social media is responsible for nearly 600,000 additional cases of depression each year in France.

The core problem we seek to address is the difficulty in identifying the specific boundary where healthy digital interactions go into excessive engagement, and so into any addiction. By focusing on Instagram's mechanics, our primary objective is to determine which demographic and behavioral factors could be the strongest predictors of a user's engagement level, to ultimately provide a clearer picture of the drivers behind platform dependency. To explore this, our study is built upon a dataset that provides the context for our analysis with over 1.5 million rows. It covers numerous distinct features ranging from demographics (age, location, income, etc.) and lifestyle habits (sleep hours, stress levels, happiness level, etc.) to detailed interaction metrics such as session frequency, time spent on various features like reels or feed, and specific engagement actions like posts created and messages sent. With such a complete dataset, we can accurately test machine learning models to uncover the hidden patterns of modern social media habits.

III. Related Work

The rise of social media addiction has emerged, with many research investigations examining its behavioural, psychological, and technological factors. A number of studies have explored the relationship between user behavior patterns and social media platforms like Facebook, Snapchat, Youtube, WeChat, WhatsApp, Instagram, etc. In the study conducted by [2], they were trying to assess the degree of social media addiction prediction through machine learning techniques and they have proved that there is a connection between satisfaction with social media and the degree of addiction, productivity loss and addiction level, as well as self-control and addiction level. In their study, K-Nearest Neighbours (KNN) appeared to be the best model (87% accuracy and ROC_AUC of 0.97) compared with Logistic Regression, Support Vector Classifier (SVC), and RF. Another similar study by [3] investigated the application of machine learning methods to identify indicators of smartphone addiction. They used three machine learning algorithms, specifically, DT, RF, and XGBoost to predict the addiction level of students to smartphone usage based on their usage patterns and user characteristics. RF achieved the best accuracy of 82.59%, followed by XGBoost (80.77%), and DT (74.56%). Their research demonstrated the effectiveness of machine learning in recognising complex patterns, which supports this project's approach of utilising a range of columns including behavioural data to predict how engaged or how addictive a user is on Instagram.

In this study, the data handling techniques employed were very similar with those used by [2], including the removal of unnecessary variables, conversion of categorical variables into numerical formats, feature scaling, and the use of feature importance ranking to identify the crucial factors influencing *user_engagement_score*. However, our approach extends this by incorporating multicollinearity checks, where variables with high Variance Inflation Factor (VIF) values (indicator of multicollinearity) were removed to eliminate redundancy.

IV. Research Methodology

a. Dataset Selection:

This study utilises a large-scale dataset that simulates Instagram user behaviour and lifestyle patterns. It contains approximately 1.5 million records and is synthetically generated to reflect realistic user activity patterns. The dataset includes a diverse set of features covering demographic, lifestyle, health, and behavioral attributes, it was selected for two main reasons. First, since it is synthetic (no real user data), it avoids privacy concerns while still preserving realistic statistical distributions and relationships between features, making it suitable for modelling behavioural patterns and is safe to use for research purposes. Second, its large size and rich set of features (58 columns) make it well-suited for distributed processing using Apache Spark. The link of the dataset is provided below.

Dataset link: <https://www.kaggle.com/datasets/rockyt07/social-media-user-analysis>

b. Exploratory Data Analysis (EDA):

Our study is based on a dataset of exactly 1,547,896 records and 58 features (38 numerical and 20 categorical) to model Instagram user behavior. No missing values or duplicate entries were found in the dataset, but a few highly-skewed features (skewness score greater than 1) were identified after conducting the skewness check. Based on the result, the highly skewed features are *followers_count* (7.97), *user_engagement_score* (3.75), *following_count* (1.51) and *sessions_per_day* (1.38).

The correlation matrix in Figure 1 shows relationships with the target variable *user_engagement_score* and highlights the features that are most strongly correlated with it. For instance, there are features that show strong negative correlations, such as *stories_viewed_per_day* (-0.562), *daily_active_minutes_instagram* (-0.530), and *time_on_feed_per_day* (-0.517). On the other hand, only *self_reported_happiness* and *age* demonstrated slight positive correlation, with values of 0.268 and 0.112 respectively.

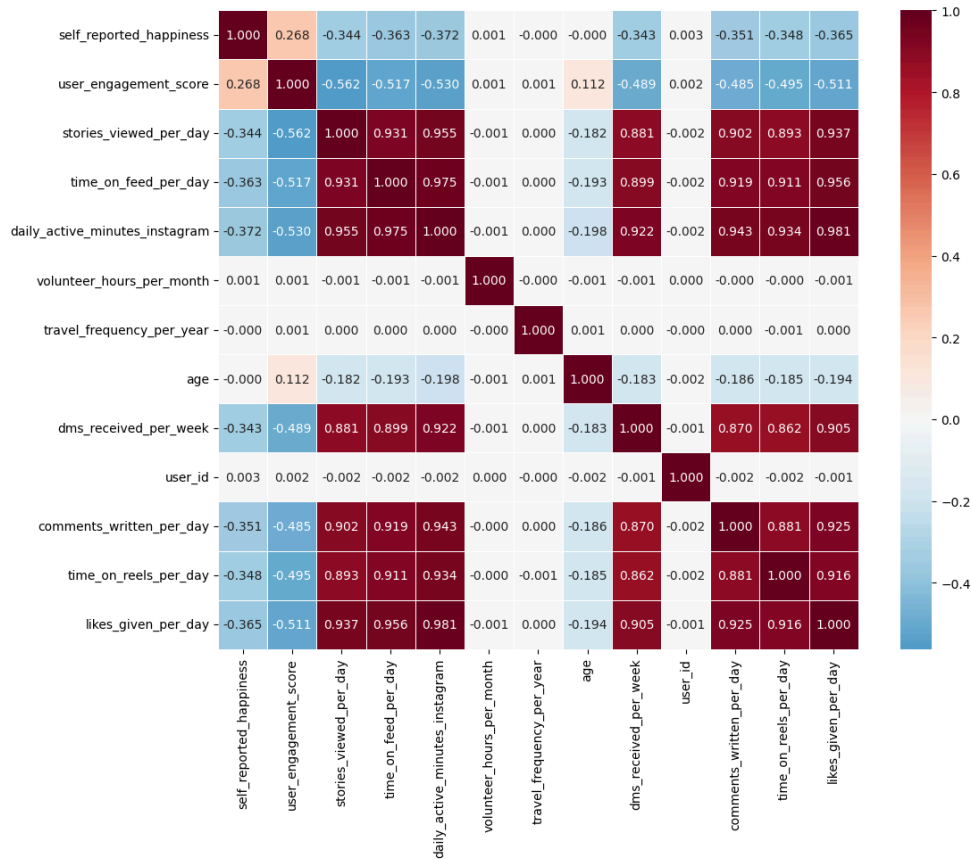


Fig. 1: Correlation Matrix with top key correlated features

Figure 2 below is a scatterplot of sessions per day against daily active minutes on instagram. From the scatterplot, it shows a strong positive correlation between the two features, which means as the daily active minutes increases, the frequency of sessions per day also tends to rise. This suggests that users who spend more time on the platform are likely opening the app more frequently throughout the day (an app's session starts when a user opens it).

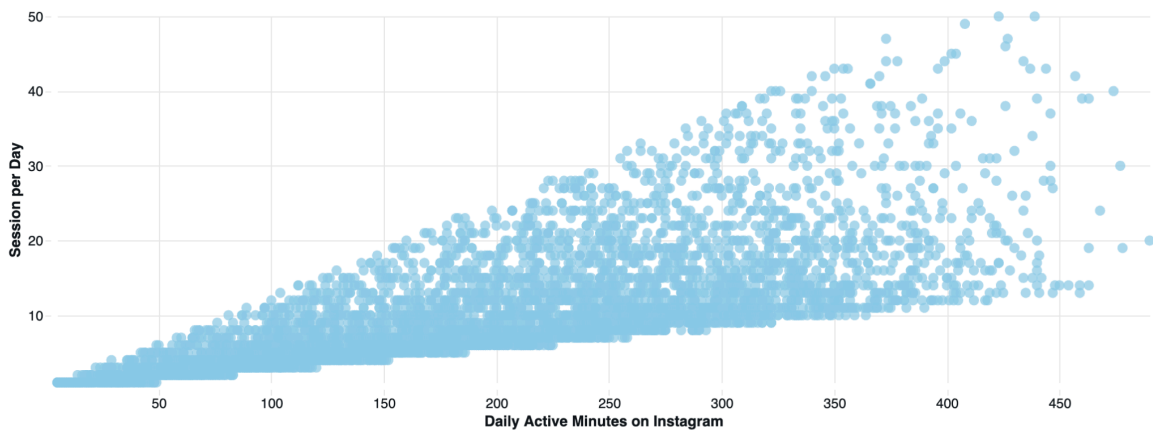


Fig. 2: Instagram Usage Time vs Sessions Frequency

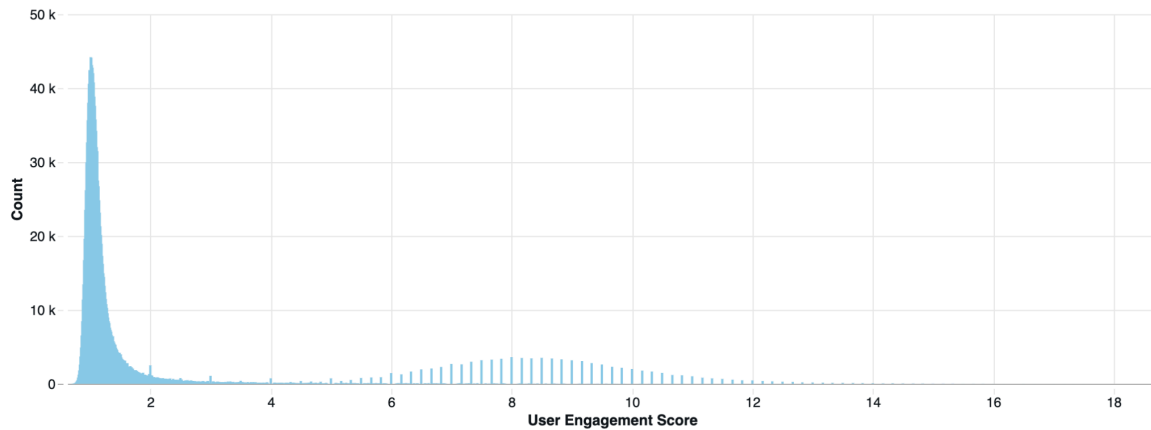


Fig. 3: User Engagement Score Distribution

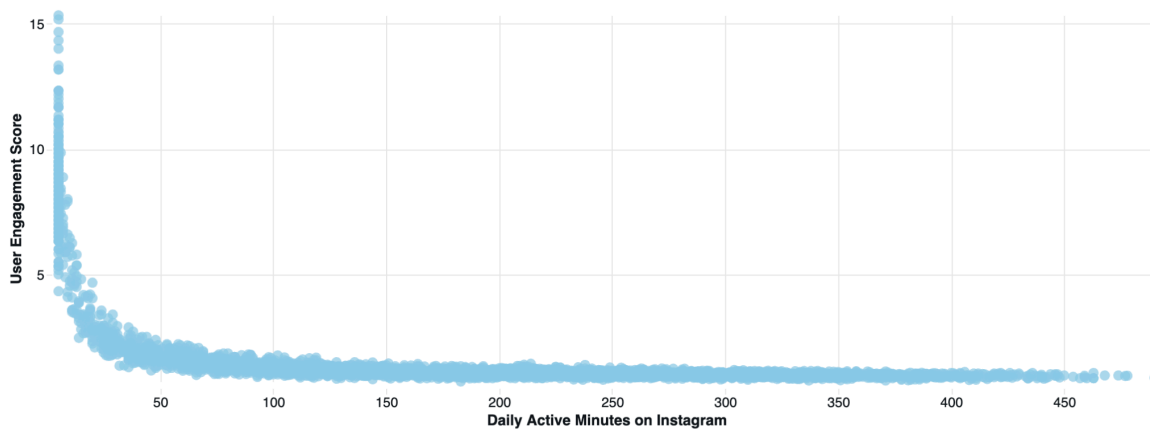


Fig. 4: Time Spent vs User Engagement Score

According to Figure 3, the distribution of the target variable, *user_engagement_score*, is highly positively skewed. To illustrate, there is a massive concentration of users at the lower end of the scale (between 0 and 2), and it has a long tail towards the right. This means there are a majority of the users maintaining a low level of active engagement but there are some users that have high engagement scores, creating an extended tail of the distribution to the right.

Figure 4 shows that users who spend the most time on the app are not necessarily the ones with the highest engagement score. In fact, peak engagement scores are primarily concentrated among users with 0 to 50 daily active minutes. This figure also illustrates an inverse relationship between usage time and levels of happiness. The graph suggests that excessive social media consumption may be associated with negative psychological outcomes, as users who tend to spend longer time on Instagram have declining values of happiness, highlighting the potential mental health risks.

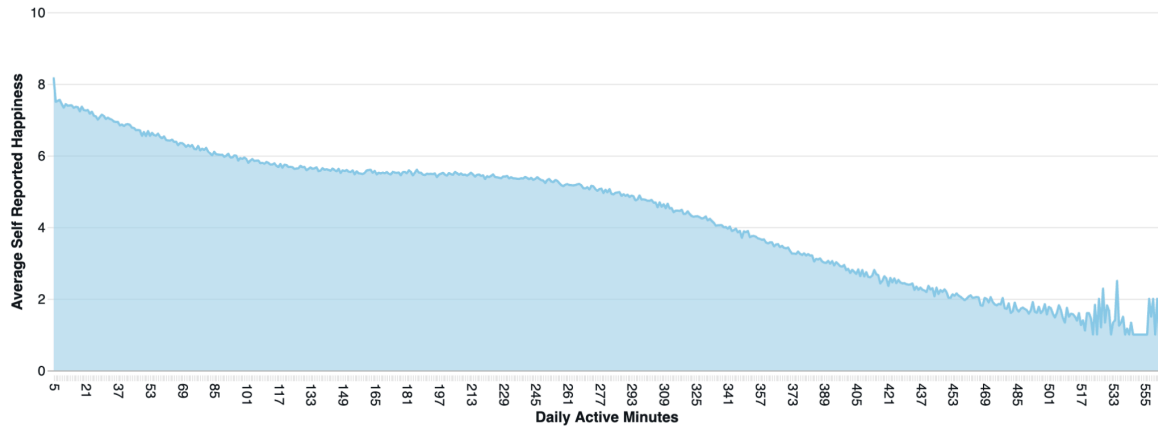


Fig. 5: Average Self Reported Happiness vs Instagram Usage

c. ML Approach:

Four models were selected in this study, namely the MLR, DT, RF, and GBT. MLR was selected as a baseline model due to its simplicity and high interpretability, making it useful for understanding linear relationships between features and the target variable [4]. In addition, tree-based models like DT, RF, and GBT, were chosen for their ability to model complex non-linear patterns. To illustrate, DT is capable of splitting data naturally into interpretable decision rules and they serve as a base learner for RF and GBT, while RF is an ensemble method that builds multiple decision trees and reduces overfitting through ensemble learning, GBT was included for their strong predictive capability, as they iteratively refine errors from previous models [4], [5], [6]. Together, these models allow for a comprehensive comparison and a more robust evaluation of performance across different modelling approaches.

A Spark ML pipeline-based approach was adopted to efficiently build and manage large-scale machine learning models. This approach enables the sequential execution of data transformation and modelling stages, ensuring scalability and consistency across different models. A VectorAssembler was utilised to combine multiple input features into a single feature vector, facilitating efficient model training. Additionally, a reusable function was used to create a modular workflow, ensuring consistency across all models.

In addition, to support model development and evaluation, the dataset was partitioned into training, validation, and testing subsets using an 80:10:10 ratio. This split ensures that model performance can be assessed reliably by only using the validation dataset to prevent data leakage during the training process. Besides, the test dataset was not used for model training or validation and it was only used after final model tuning, ensuring an unbiased evaluation of the final model.

Apache Spark was used to enable distributed data processing and scalable machine learning. All transformations were performed using Spark DataFrames, and models were implemented using SparkML pipelines to ensure efficiency when handling large-scale data.

d. Data Preprocessing & Feature Engineering:

The dataset does not contain any missing values in any of the columns or duplicated rows, which is a good starting point. However, several cleaning and transformation steps were still necessary to make the dataset more suitable for modelling. First, columns that do not contribute meaningful information like *user_id*, *app_name*, and *last_login_date* were removed. The *user_id* acts as a unique identifier and has no predictive value for estimating the *user_engagement_score*. Similarly, *app_name* contains only a single constant value called Instagram, so it does not provide any variation for the model to learn from. As for *last_login_date*, although it is a time-related feature, keeping it in its raw form is not useful and provides no meaningful interpretation to the model. While it could be transformed into a new feature like “*days_since_last_login*”, the data shows that most users have not logged in for a long period (ranging from at least four months to over a year), which means they are not active and not engaging, therefore, removing this column helps simplify the model and would be a better choice instead of transforming it.

Next, data transformation was applied on a column called *account_creation_year* into a new column called *account_age* by subtracting the values with the current year 2026. This allows the model to better capture how long a user has been on the platform, which is more interpretable than a raw year value.

Binary categorical columns with values like “yes” and “no” were converted into numerical format (1 and 0), as regression models can only work with numeric inputs. Besides, categorical variables were one-hot encoded as well so that the model can properly interpret them.

For numerical features, columns with highly-skewed distributions were log-transformed to reduce skewness and make the data more normally distributed. Columns with large differences in scale such as *followers_count* (one can have only a few followers and some may have many followers) and *following_count* were standardised using z-score scaling to ensure that no single feature dominates the model due to its magnitude.

Lastly, multicollinearity was checked using VIF and features with high VIF values were removed to prevent redundancy and to ensure that the model only learns independent relationships between variables.

On the other hand, for feature engineering, the full set of available features were included during the initial model training phase, to provide a baseline benchmark for comparison with later models built using a reduced feature set. Feature selection was

carried out based on feature importance derived from tree-based models. Although the importance rankings from the DT, RF, and GBT were not exactly the same across all features, the top eight features were consistent among three of them and were therefore selected. On top of that, three more features were included, namely the *perceived_stress_score*, *self_reported_happiness*, and *ads_clicked_per_day*. The first two were included as they represent mental health-related behaviours and differ from the initial eight features, which are all platform-based interactions metrics. Together with *ads_clicked_per_day*, their importance values further support their inclusion in the feature set (on the order 10^{-4}). The remaining features were excluded as their importance values were much lower (around 10^{-6}), indicating minimal contribution to the model's predictive performance. In short, based on feature importance, 11 features were initially selected, but the final set was reduced to 10 after removing *average_session_length_minutes*, as it showed a high correlation with *sessions_per_day* and the target variable, *user_engagement_score*.

This reduced feature set approach was taken to eliminate redundant or less significant variables, thereby reducing model complexity and improving model efficiency while maintaining predictive performance. Additionally, the reduced feature set enhances model interpretability by focusing only on the most impactful behavioural attributes.

V. Evaluation

a. Model Performance Evaluation:

For this regression task, model performance was evaluated using R^2 and RMSE, as classification metrics are not applicable. R^2 measures the proportion of variance explained and RMSE captures error magnitude (penalising large errors).

A_C^B model	1.2 train_r2	1.2 val_r2	1.2 train_rmse	1.2 val_rmse
Linear Regression	0.94325668455558...	0.94288989602619	0.12670781970329417	0.12699578040461332
Decision Tree	0.94858523944591...	0.94851009826966...	0.12061184835165042	0.12058515536599654
Random Forest	0.96137720684174...	0.96105289179568...	0.10453648748459005	0.10487462582883555
GBT	0.98268862311308...	0.98251034302209...	0.06998611240652151	0.07027865829589576

Table 1: Initial results for full features and top 11 features

The Table 1 above shows the results of the four models using the full set of features and top 11 features. All models achieved nearly similar train and validation results, which showed that there is no overfitting issue. MLR and DT achieved around 0.94 R^2 and 0.12 RMSE, RF achieved 0.96 R^2 and 0.10 RMSE, while GBT achieved the highest R^2 (~ 0.98) and the lowest RMSE (0.07).

b. Case Study and Scenario Testing:

To validate real-world applicability, three representative user scenarios were tested using the tuned GBT model:

- Heavy user: High sessions, interactions, low happiness → predicted high engagement (~8.76);
- Casual user: Low activity, high happiness → predicted low engagement (~1.72);
- Stress-driven user: Moderate activity, high stress → predicted moderate engagement (~4.18).

Scenario	User Type	Predicted Score (Raw)	Predicted Score (Log)	Aligned with expectation?
1	Heavy / Addicted User	~8.76	~2.17	Yes
2	Casual / Healthy User	~1.72	~0.54	Yes
3	Stress-Driven User	~4.18	~1.43	Yes

Table 2: Scenario Testing Summary - GBT Predictions

All predictions explained in Table 2 aligned with expected behavioural patterns, indicating that the model captures both behavioural and psychological signals effectively and generalises beyond aggregate metrics.

c. Comparative Analysis - State-of-the-Art and Baseline Benchmarking:

As GBT is the best performed model, it was compared against a relevant study by [2]. In their study, they used KNN and they achieved an accuracy of 87%, while our model achieved an R^2 value of 0.90.

Author	Platform	Task	Best Model	Best Performance	Dataset
Alade et al. (2026)	Multi-platform	Classification	KNN	87% accuracy	2000 rows and 28 features
Our Study	Instagram	Regression	GBT	R^2 - 0.90	1.5 million rows and 58 features

Table 3: Summary Comparison — State-of-the-Art vs Our Study

VI. Results

a. Key Results and Visualisations:

The final model comparison in Table 4 shows that GBT consistently outperformed all models, confirming that ensemble methods are best suited for this problem.

Model	Train R ²	Validation R ²	Train RMSE	Validation RMSE
GBT	0.9049	0.9032	0.1640	0.1660
RF	0.8926	0.8914	0.1742	0.1750
DT	0.8660	0.8653	0.1947	0.1751
LR	0.6338	0.6338	0.3219	0.3216

Table 4: Final Model Performance Comparison (Training vs Validation)

Model	1.2 Val R2	1.2 Val RMSE	1.2 Test R2	1.2 Test RMSE
Gradient Boosted Trees	0.9032	0.1654	0.9038	0.1644

Table 5: GBT Final evaluation on Test dataset

The table 5 above is the result of GBT being evaluated on the unseen test dataset, and it can be seen that the model performed well as its results improved from the validation dataset results. This indicates that the model is generalising well and not memorising the data.

The histogram below in Fig. 6 shows a normal distribution, this indicates that the errors are tightly clustered around 0, showing that the model's predictions are centrally balanced. Furthermore, the high density near zero also suggests that the GBT model has effectively captured the variance in the dataset.

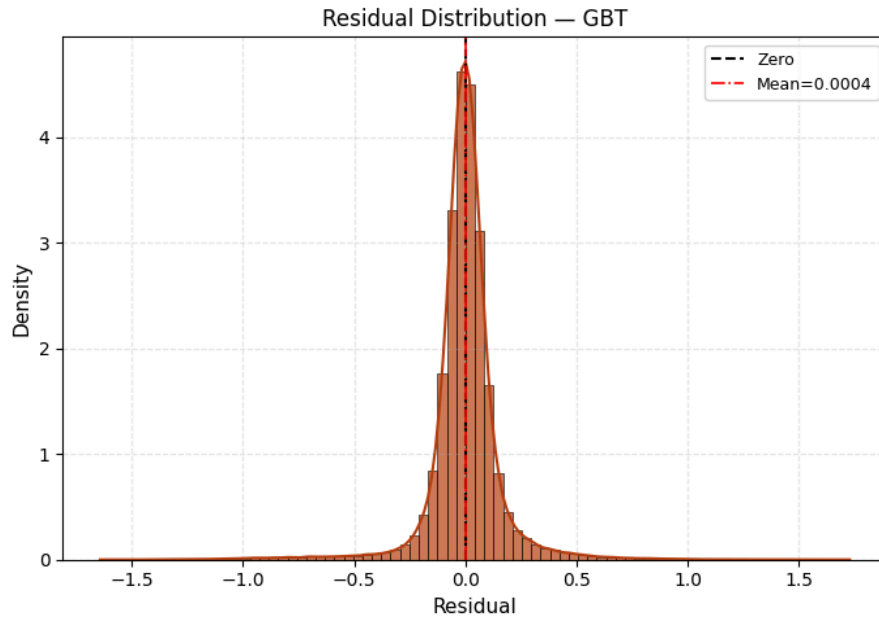


Fig. 6: Residual Distribution - GBT

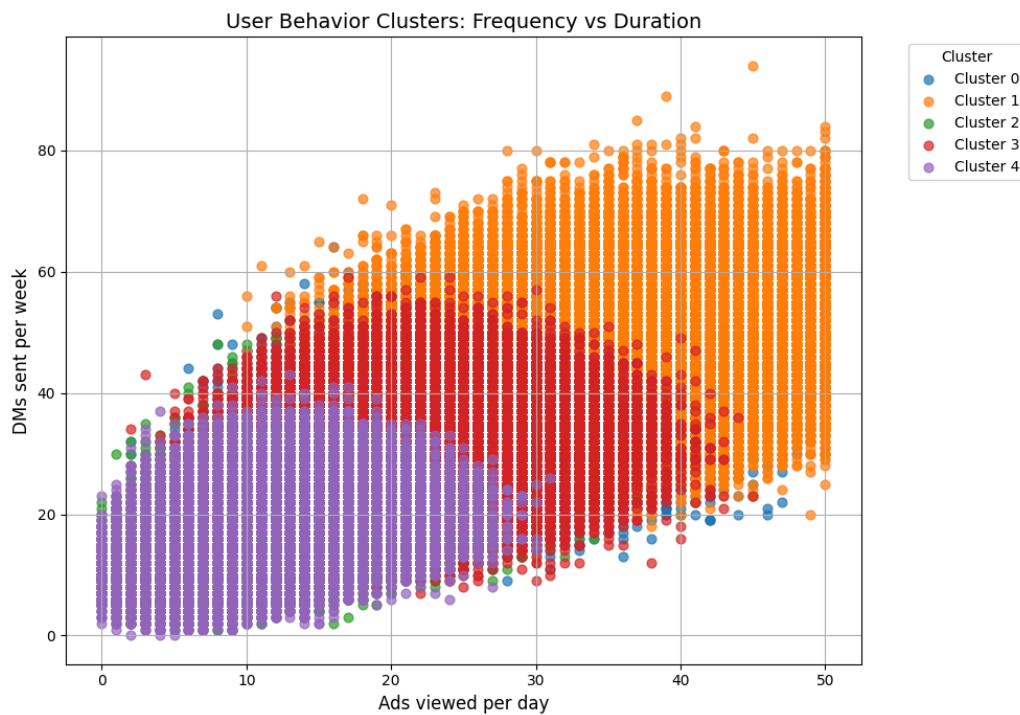


Fig. 7: User Behavior Clusters

The above scatter plot represents the relationship between ads viewed per day and DMs sent per week across five distinct user clusters identified through KMeans clustering. Cluster 1(Orange) stands out as the most active segment, comprising users who are exposed to the highest number of ads and simultaneously engage in the most direct messaging activity, suggesting a highly interactive user profile. In contrast, Cluster 4(Purple)

represents the largest and most restrained group.

Summary of Best Model (GBT)

Metric	Value
Model	Gradient Boosted Trees
Configuration	maxDepth=5, maxIter=100, stepSize=0.1
Test R ²	0.9038
Test RMSE	0.1644
Most Important Feature	<i>sessions_per_day</i>
Least Important Feature	<i>perceived_stress_score</i>

Table 6: Summary of Best Performing Model (Gradient Boosted Trees)

VII. Discussion

One of the primary challenges encountered in this study was the dominance of a small subset of behavioural interaction features, where a few variables contributed disproportionately to the model predictions. This resulted in the model relying heavily on platform-based interaction attributes, while demographic, lifestyle, and health-related features had minimal influence. Such dominance can reduce model interpretability and limit the ability to understand broader patterns within the data. To address this issue, feature reduction techniques were applied by removing highly dominant features, allowing the model to consider a more balanced set of variables.

Additionally, significant multicollinearity was observed among several behavioural features, particularly interaction-related variables such as *daily_active_minutes_instagram*, *likes_given_per_day*, and *time_on_feed_per_day*, which exhibited very high VIF values. This indicates a strong linear dependency among these variables, leading to redundancy in the feature space and potentially unstable model behaviour. To solve this, features with high VIF values were manually removed, thereby reducing redundancy and improving model stability.

Another challenge encountered was during the hyperparameter tuning process within the Databricks environment. For instance, since Databricks lacks built-in automated cross-validation functions, a manual implementation was required. Also, because the dataset was so large, 5-fold cross-validation took too much time to run and therefore, a 3-fold cross-validation approach was taken instead, to balance speed and performance.

The most significant limitation of this project is its restricted platform scope, as the analysis is based exclusively on Instagram data and does not include any data from other social media platforms. Besides, while the removal of dominant features and high-VIF variables is necessary to stabilise the model, it may reduce the richness of the dataset and potentially cap the model's maximum performance.

Additionally, Spark-based distributed processing was used to efficiently handle the large-scale dataset.

VIII. Conclusion

The main focus of this project was to see how accurately a user's engagement level on Instagram can be predicted using MLR, DT, RF, and GBT, as well as what are the key factors that greatly contribute to it. Among the four models, GBT performed best, achieving the lowest RMSE of approximately 0.1654 and the highest R^2 score of approximately 0.9032 among all models. Besides, by leveraging the feature importance analysis inherent in tree-based models, the project successfully identified and ranked the most significant factors, highlighting a top 10 list that consists of eight platform-based metrics and two mental health-related variables.

In comparison with existing studies, the proposed model demonstrates stronger predictive performance, achieving an R^2 of approximately 0.9032, which exceeds the accuracy levels reported in related classification-based approaches (typically ranging from 82% to 87%). This improvement can be attributed to the larger dataset size, platform-specific feature set, and robust preprocessing techniques applied in this study.

Despite these results, several areas could benefit from further improvement. First, to improve model robustness, future studies should utilise even larger datasets. While the models were trained on a dataset with around 1.5 million rows and 58 features in this project, increasing the volume and variety of data would likely yield deeper insights. Second, expanding the scope to include data from other social media platforms beyond Instagram would allow for a more comprehensive cross-platform analysis, as this project only includes Instagram data. Third, the hyperparameter optimisation process was limited in scope because the tuning of the models did not include a wide range of parameters, therefore, more optimal configurations may have been overlooked. Finally, while this study focused on MLR and tree-based approaches, incorporating a wider variety of models, such as neural networks, could provide a more comprehensive comparative analysis and potentially enhance predictive performance.

IX. References

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